RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 144/194 fusions (74.2%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=9.84123e-05, statistically significant at alpha=0.05). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 74.2% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 119/146 (81.5%) of in-frame fusions retained the domain, p=9.84123e-05 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00699301.

Domain disruption

Domain-disruption analysis was skipped: RET has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1063 aa (permutation-corrected p=0.000999001, statistically significant at alpha=0.05). This is 58 aa from the nearest configured domain boundary at 1005 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: Algorithm failed: ValueError: params['group_field'] is required

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: Algorithm failed: ValueError: joint_partner requires a GeneConfig with gene_pair

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	87	0.4484536082474227	0.21553360374650618	None	0.6565794315648202	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.40986628306510914	1
CCDC6	51	0.26288659793814434	0.21553360374650618	None	0.7873416786677006	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.35835804588317843	2
NCOA4	16	0.08247422680412371	0.21553360374650618	None	0.8632886064030132	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2973127999118795	3
FBXL7	2	0.010309278350515464	0.21553360374650618	None	1.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2956802634882588	4
CTNNA1	1	0.005154639175257732	0.21553360374650618	None	0.9585687382297552	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28459300489095307	5
RUFY2	1	0.005154639175257732	0.21553360374650618	None	0.9510357815442562	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28297879988691754	6
CTNNA3	1	0.005154639175257732	0.21553360374650618	None	0.8785310734463276	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26744207672307574	7
RUFY1	2	0.010309278350515464	0.21553360374650618	None	0.7871939736346516	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2500789721242556	8
SPECC1L	4	0.020618556701030927	0.21553360374650618	None	0.6697269303201507	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22932572499279777	9
ERCC6	3	0.015463917525773196	0.21553360374650618	None	0.6704331450094162	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22726792563681564	10
TIMM23B	3	0.015463917525773196	0.21553360374650618	None	0.6701192718141871	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2272006670949808	11
RUFY3	2	0.010309278350515464	0.21553360374650618	None	0.6694915254237288	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22485701893620072	12
CLIP1	1	0.005154639175257732	0.21553360374650618	None	0.6704331450094162	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22284966348659468	13
GRIPAP1	1	0.005154639175257732	0.21553360374650618	None	0.6704331450094162	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22284966348659468	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RASSF4	1	0.005154639175257732	0.21553360374650618	None	0.6704331450094162	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22284966348659468	15
SHROOM3	1	0.005154639175257732	0.21553360374650618	None	0.6704331450094162	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22284966348659468	16
TRIM24	1	0.005154639175257732	0.21553360374650618	None	0.6704331450094162	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22284966348659468	17
CCDC186	1	0.005154639175257732	0.21553360374650618	None	0.6694915254237288	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22264788786109027	18
CUBN	1	0.005154639175257732	0.21553360374650618	None	0.6694915254237288	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22264788786109027	19
GEMIN5	1	0.005154639175257732	0.21553360374650618	None	0.6694915254237288	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22264788786109027	20
RELCH	1	0.005154639175257732	0.21553360374650618	None	0.6694915254237288	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22264788786109027	21
SLC4A4	1	0.005154639175257732	0.21553360374650618	None	0.6694915254237288	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22264788786109027	22
EML4	1	0.005154639175257732	0.21553360374650618	None	0.6327683615819208	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.21477863846641712	23
RBPMS	1	0.005154639175257732	0.21553360374650618	None	0.6177024482109228	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.21155022845834615	24
ERC1	1	0.005154639175257732	0.21553360374650618	None	0.60075329566855	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2079182671992662	25
PDCD10	1	0.005154639175257732	0.21553360374650618	None	0.5894538606403013	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.20549695969321294	26
TFG	1	0.005154639175257732	0.21553360374650618	None	0.5894538606403013	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.20549695969321294	27
LIN52	1	0.005154639175257732	0.21553360374650618	None	0.551789077212806	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.19742593467303537	28
DOCK1	1	0.005154639175257732	0.21553360374650618	None	0.5225988700564972	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.19117089028239775	29
KCNMA1	1	0.005154639175257732	0.21553360374650618	None	0.5169491525423728	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1899602365293711	30

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1217	1	0.005154639175257732	0.21553360374650618	None	0.39924670433145004	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16473828334131624	31
UXS1	1	0.005154639175257732	0.21553360374650618	None	0.20527306967984937	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.12317250448740183	32
KIF13A	1	0.005154639175257732	0.21553360374650618	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.0791854181274341	33

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1	1	143	1	49	0.34265734265734266	0.45002937877250143	0.34675913373064243
219	1	143	2	48	0.16783216783216784	0.16358634688317933	0.7862529459014297
425	2	142	2	48	0.3380281690140845	0.2738144245997509	0.5625436768774693
550	4	140	3	47	0.44761904761904764	0.37669413920559647	0.424011136713296
556	5	139	3	47	0.5635491606714629	0.4269916226168348	0.369580645553364
587	6	138	4	46	0.5	0.28474795225204086	0.5455393904405627
622	7	137	4	46	0.5875912408759124	0.47853605021856643	0.32008533934743655
627	8	136	6	44	0.43137254901960786	0.19970190836413992	0.699617784963772
639	9	135	6	44	0.4888888888888889	0.22050970732209507	0.6565722871774424
657	10	134	6	44	0.5472636815920398	0.3689676566450757	0.43301170201493716
663	11	133	6	44	0.606516290726817	0.3861986635634364	0.4131892334256796
671	13	131	6	44	0.727735368956743	0.5831276940464237	0.2342363324669072
673	14	130	6	44	0.7897435897435897	0.6010603140306741	0.22108194607309356
687	15	129	6	44	0.8527131782945736	0.7931139769308365	0.1006643965471982
708	16	128	6	44	0.9166666666666666	0.8019071605022902	0.09587590854306939
712	74	70	8	42	5.55	1.0784500183907644e-05	4.967199977812329
713	141	3	14	36	120.85714285714286	3.386108118610997e-24	23.470299178959955
934	141	3	15	35	109.66666666666667	3.253299303362719e-23	22.487675979810888
1019	142	2	15	35	165.66666666666666	2.831240208694676e-24	23.548023282565293
1063	144	0	15	35	None	5.1561797693606186e-27	26.287671949448907

domain_retention tables

frame_domain_contingency_table

119	25
27	23

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CCDC186	1
CCDC6	51
CLIP1	1
CTNNA1	1
CTNNA3	1
CUBN	1
DOCK1	1
EML4	1
ERC1	1
ERCC6	3
FBXL7	2
GEMIN5	1
GRIPAP1	1
KCNMA1	1
KIAA1217	1
KIF13A	1
KIF5B	87
LIN52	1
NCOA4	16
PDCD10	1
RASSF4	1
RBPMS	1
RELCH	1
RUFY1	2
RUFY2	1
RUFY3	2
SHROOM3	1
SLC4A4	1

Partner_gene	Event_count
SPECC1L	4
TFG	1
TIMM23B	3
TRIM24	1
UXS1	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051781-T03-IM6-3	P-0051781-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61651178	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0059857-T02-IM7-4	P-0059857-T02-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	61604308	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0065992-T01-IM7-5	P-0065992-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	61653659	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0049461-T02-IM6-6	P-0049461-T02-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61615910	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0036379-T01-IM6-7	P-0036379-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61639380	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0037412-T01-IM6-8	P-0037412-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61624434	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0048611-T01-IM6-9	P-0048611-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61652942	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064853-T01-IM7-10	P-0064853-T01-IM7		CCDC6-RET	CCDC6	out-of-frame	three_prime	61629044	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv | Protein Fusion: out of frame {CCDC6:RET}	NA
EVT-P-0047144-T01-IM6-11	P-0047144-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61657033	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0038257-T01-IM6-12	P-0038257-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61614591	20	1115	True	lost		Protein kinase domain		CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12):c.304-2131:CCDC6_c.2136+410:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0017365-T04-IM6-13	P-0017365-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61640379	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0039548-T01-IM6-14	P-0039548-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61621961	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0040111-T01-IM6-15	P-0040111-T01-IM6		CCDC6-RET	CCDC6	unknown	three_prime	61626333	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv | Protein Fusion: mid-exon {CCDC6:RET}	NA
EVT-P-0041192-T01-IM6-16	P-0041192-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61657248	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T04-IM6-17	P-0022642-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61626673	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T03-IM6-18	P-0022642-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61626673	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0051781-T01-IM6-19	P-0051781-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61651178	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042417-T01-IM6-20	P-0042417-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61649492	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0004901-T01-IM5-21	P-0004901-T01-IM5		CCDC6-RET	CCDC6	in-frame	three_prime	61653032	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0022380-T03-IM6-22	P-0022380-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61644070	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0068500-T01-IM7-23	P-0068500-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	61654590	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0057735-T01-IM6-24	P-0057735-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	61665755	20	1115	True	lost		Protein kinase domain		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0038053-T01-IM6-26	P-0038053-T01-IM6		CLIP1-RET	CLIP1	unknown	three_prime	43611121	12	713	True	retained	Protein kinase domain			CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512) | Protein Fusion: mid-exon {CLIP1:RET}	NA
EVT-P-0000252-T01-IM3-27	P-0000252-T01-IM3		CUBN-RET	CUBN	unknown	three_prime	43610494	11	712	True	retained	Protein kinase domain			NA | Antisense fusion	NA
EVT-P-0038256-T01-IM6-28	P-0038256-T01-IM6		ERC1-RET	ERC1	unknown	three_prime	43609964	11	639	False	retained	Protein kinase domain			ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1; p13.32)(chr10:g.43609964::chr12:g.1264106) | Protein Fusion: mid-exon {ERC1:RET}	NA
EVT-P-0034781-T01-IM6-29	P-0034781-T01-IM6		GRIPAP1-RET	GRIPAP1	in-frame	three_prime	43611853	12	713	True	retained	Protein kinase domain			GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853) | Protein Fusion: in frame {GRIPAP1:RET}	NA
EVT-P-0026245-T01-IM6-30	P-0026245-T01-IM6		KIAA1217-RET	KIAA1217	unknown	three_prime	43606666	7	425	False	retained	Protein kinase domain			KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel | Protein Fusion: mid-exon {KIAA1217:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042153-T01-IM6-31	P-0042153-T01-IM6		KIF13A-RET	KIF13A	in-frame	three_prime	17806558	1	1	True	retained	Protein kinase domain			KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510) | Protein Fusion: in frame {KIF13A:RET}	NA
EVT-P-0019535-T02-IM6-32	P-0019535-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611913	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039257-T03-IM7-33	P-0039257-T03-IM7		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0009514-T01-IM5-34	P-0009514-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611811	12	713	True	retained	Protein kinase domain			Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264 | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0055183-T01-IM6-35	P-0055183-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0011035-T01-IM5-36	P-0011035-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611322	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20): c.1726-2296:KIF5B_c.2137-710:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0012648-T01-IM5-37	P-0012648-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611697	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0053175-T01-IM6-38	P-0053175-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611162	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0052861-T02-IM6-39	P-0052861-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610260	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013825-T01-IM5-40	P-0013825-T01-IM5		KIF5B-RET	KIF5B	unknown	three_prime	43611520	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv | Protein fusion: mid-exon (KIF5B-RET)	NA
EVT-P-0052826-T01-IM6-41	P-0052826-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611182	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T01-IM6-42	P-0014808-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611055	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0034357-T01-IM6-43	P-0034357-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611187	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T02-IM6-44	P-0014808-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0033700-T02-IM6-45	P-0033700-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611670	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T03-IM6-46	P-0014808-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0008289-T01-IM5-47	P-0008289-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611748	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0000252-T01-IM3-48	P-0000252-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43610597	11	712	True	retained	Protein kinase domain			RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv) | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008101-T01-IM5-49	P-0008101-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610455	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0048968-T01-IM6-50	P-0048968-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611809	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0048829-T01-IM6-51	P-0048829-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610616	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0065876-T01-IM7-52	P-0065876-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611299	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0016221-T01-IM6-53	P-0016221-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0037367-T01-IM6-54	P-0037367-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611874	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0057072-T02-IM6-55	P-0057072-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611106	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0017214-T01-IM6-56	P-0017214-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610647	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0017214-T02-IM6-57	P-0017214-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610647	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032753-T01-IM6-58	P-0032753-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610925	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0057072-T03-IM7-59	P-0057072-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611106	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032686-T01-IM6-60	P-0032686-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611982	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0017523-T02-IM6-61	P-0017523-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611192	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046827-T02-IM6-62	P-0046827-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610889	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032407-T01-IM6-63	P-0032407-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611730	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0018069-T01-IM6-64	P-0018069-T01-IM6		KIF5B-RET	KIF5B	out-of-frame	five_prime	43607832	8	550	True	lost		Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv | Protein Fusion: out of frame {RET:KIF5B}	NA
EVT-P-0005573-T01-IM5-65	P-0005573-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610418	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0004995-T01-IM5-66	P-0004995-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610660	11	712	True	retained	Protein kinase domain			KIF5B(NM_004521) -RET (NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036092-T01-IM6-67	P-0036092-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610413	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046726-T01-IM6-68	P-0046726-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610987	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032227-T01-IM6-69	P-0032227-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611125	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0020273-T02-IM6-70	P-0020273-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611819	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T02-IM6-71	P-0043007-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610813	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021150-T01-IM6-72	P-0021150-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611591	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021578-T01-IM6-73	P-0021578-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610809	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021879-T01-IM6-74	P-0021879-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610699	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021937-T01-IM6-75	P-0021937-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021937-T03-IM6-76	P-0021937-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T01-IM6-77	P-0043007-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610813	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042608-T01-IM6-78	P-0042608-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611802	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042580-T01-IM6-79	P-0042580-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610944	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0059913-T01-IM7-80	P-0059913-T01-IM7		KIF5B-RET	KIF5B	unknown	three_prime	43610108	11	687	False	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0001010-T02-IM5-81	P-0001010-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610863	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0060748-T02-IM7-82	P-0060748-T02-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042104-T01-IM6-83	P-0042104-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610505	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0003125-T01-IM5-84	P-0003125-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611584	12	713	True	retained	Protein kinase domain			NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0003125-T01-IM5-85	P-0003125-T01-IM5		KIF5B-RET	KIF5B	in-frame	five_prime	43611581	12	713	True	lost		Protein kinase domain		NA | Protein fusion: in frame (RET-KIF5B)	NA
EVT-P-0002873-T01-IM3-86	P-0002873-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43611244	12	713	True	retained	Protein kinase domain			NA | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022866-T01-IM6-87	P-0022866-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611871	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0024225-T01-IM6-88	P-0024225-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43609110	10	622	False	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0002694-T02-IM5-89	P-0002694-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611682	12	713	True	retained	Protein kinase domain			NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0024843-T01-IM6-90	P-0024843-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611733	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025100-T01-IM6-91	P-0025100-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025100-T02-IM6-92	P-0025100-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039941-T01-IM6-93	P-0039941-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611339	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0060748-T03-IM7-94	P-0060748-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039474-T02-IM6-95	P-0039474-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610359	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039474-T01-IM6-96	P-0039474-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610359	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055183-T02-IM6-97	P-0055183-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025838-T01-IM6-98	P-0025838-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610972	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025928-T01-IM6-99	P-0025928-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025928-T02-IM6-100	P-0025928-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0036637-T01-IM6-101	P-0036637-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039257-T01-IM6-102	P-0039257-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0026693-T01-IM6-103	P-0026693-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610511	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5_c.2136+327:RETinv. | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026777-T01-IM6-104	P-0026777-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026777-T02-IM6-105	P-0026777-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0038961-T01-IM6-106	P-0038961-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610775	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027723-T01-IM6-107	P-0027723-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20):c.1725+966:KIF5B_c.2123:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0036737-T02-IM6-108	P-0036737-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43612025	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0061331-T01-IM7-109	P-0061331-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610884	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0027991-T01-IM6-110	P-0027991-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43608201	9	550	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0062948-T03-IM7-111	P-0062948-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610211	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0064951-T01-IM7-112	P-0064951-T01-IM7		KIF5B-RET	KIF5B	out-of-frame	three_prime	43611130	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv | Protein Fusion: out of frame {KIF5B:RET}	NA
EVT-P-0030453-T01-IM6-113	P-0030453-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610911	11	712	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0037494-T01-IM6-114	P-0037494-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611755	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0000208-T02-IM5-115	P-0000208-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20): c.1726-1971:KIF5B_c.2137-385:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0000208-T01-IM3-116	P-0000208-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	Protein kinase domain			RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016221-T02-IM6-117	P-0016221-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	retained	Protein kinase domain			KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0019663-T01-IM6-118	P-0019663-T01-IM6		LIN52-RET	LIN52	out-of-frame	five_prime	43608613	9	587	True	lost		Protein kinase domain		RET (NM_020975) - LIN52 (NM_001024674) Rearrangement : t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750) | Protein Fusion: out of frame {RET:LIN52}	NA
EVT-P-0065826-T01-IM7-120	P-0065826-T01-IM7		NCOA4-RET	NCOA4	out-of-frame	three_prime	51587635	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0037961-T01-IM6-121	P-0037961-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	51584256	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0038315-T01-IM6-122	P-0038315-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	51584201	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0060765-T02-IM7-123	P-0060765-T02-IM7		NCOA4-RET	NCOA4	out-of-frame	three_prime	51584168	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0043077-T01-IM6-124	P-0043077-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	51583547	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0005748-T01-IM5-125	P-0005748-T01-IM5		NCOA4-RET	NCOA4	in-frame	three_prime	51584356	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0047991-T02-IM6-126	P-0047991-T02-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	51583476	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0048805-T01-IM6-127	P-0048805-T01-IM6		NCOA4-RET	NCOA4	unknown	three_prime	51584704	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049600-T01-IM6-128	P-0049600-T01-IM6		NCOA4-RET	NCOA4	unknown	three_prime	51586611	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0056870-T01-IM6-129	P-0056870-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	51582404	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0052409-T01-IM6-130	P-0052409-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	51582486	20	1115	True	lost		Protein kinase domain		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0004094-T02-IM6-131	P-0004094-T02-IM6		RELCH-RET	RELCH	in-frame	three_prime	43610279	11	712	True	retained	Protein kinase domain			KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.21;q21.33)(chr10:g.43610279::chr18:g.59909399) | Protein Fusion: in frame {KIAA1468:RET}	NA
EVT-P-0015588-T01-IM6-132	P-0015588-T01-IM6		RET-CCDC186	CCDC186	in-frame	three_prime	43610308	11	712	True	retained	Protein kinase domain			C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142:C10orf118_c.2136+124:RETinv | Protein fusion: in frame (C10orf118-RET)	NA
EVT-P-0020974-T01-IM6-133	P-0020974-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611184	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-848:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0001745-T02-IM5-134	P-0001745-T02-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611245	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0017365-T02-IM6-135	P-0017365-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0001898-T01-IM3-136	P-0001898-T01-IM3		RET-CCDC6	CCDC6	in-frame	three_prime	43611699	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029602-T01-IM6-137	P-0029602-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611160	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0004916-T01-IM5-138	P-0004916-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	Protein kinase domain			CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0000425-T01-IM3-139	P-0000425-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43610450	11	712	True	lost		Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) | Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-0031330-T01-IM6-140	P-0031330-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0030990-T03-IM6-141	P-0030990-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611881	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0005168-T01-IM5-142	P-0005168-T01-IM5		RET-CCDC6	CCDC6	out-of-frame	three_prime	43608797	10	587	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv | Protein fusion: out of frame (CCDC6-RET)	NA
EVT-P-0022380-T01-IM6-143	P-0022380-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022380-T02-IM6-144	P-0022380-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2136+475inv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0029315-T02-IM6-145	P-0029315-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029315-T01-IM6-146	P-0029315-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0000889-T01-IM3-147	P-0000889-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43611994	12	713	True	lost		Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv | Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-0007322-T02-IM6-148	P-0007322-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610268	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0027953-T01-IM6-149	P-0027953-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611278	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0008504-T03-IM6-150	P-0008504-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610583	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0008955-T03-IM5-151	P-0008955-T03-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611019	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0013023-T01-IM5-152	P-0013023-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611810	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0013394-T01-IM5-153	P-0013394-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610232	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0022607-T01-IM6-154	P-0022607-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611212	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022642-T01-IM6-155	P-0022642-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	Protein kinase domain			CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T02-IM6-156	P-0022642-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0017365-T03-IM6-157	P-0017365-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0025349-T03-IM6-158	P-0025349-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611986	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0049588-T01-IM6-159	P-0049588-T01-IM6		RET-CCDC6	CCDC6	out-of-frame	five_prime	43609437	10	627	True	lost		Protein kinase domain		RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv | Protein Fusion: out of frame {RET:CCDC6}	NA
EVT-P-0015435-T01-IM6-160	P-0015435-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610969	11	712	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0005556-T01-IM5-161	P-0005556-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611828	12	713	True	retained	Protein kinase domain			CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0063502-T01-IM7-162	P-0063502-T01-IM7		RET-CTNNA1	CTNNA1	unknown	five_prime	43622038	19	1019	False	retained	Protein kinase domain			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038) | Protein Fusion: mid-exon {RET:CTNNA1}	NA
EVT-P-0063802-T01-IM7-163	P-0063802-T01-IM7		RET-CTNNA3	CTNNA3	out-of-frame	five_prime	43617566	16	934	True	disrupted			Protein kinase domain	RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-34298:CTNNA3inv | Protein Fusion: out of frame {RET:CTNNA3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030396-T02-IM6-164	P-0030396-T02-IM6		RET-DOCK1	DOCK1	unknown	three_prime	43608320	9	556	False	retained	Protein kinase domain			DOCK1 (NM_001380) - RET (NM_020975) Rearrangement: c.2515+295:RET_c.1668:RETdup | Protein Fusion: mid-exon {DOCK1:RET}	NA
EVT-P-0013886-T01-IM5-165	P-0013886-T01-IM5		RET-EML4	EML4	unknown	three_prime	43610067	11	673	False	retained	Protein kinase domain			EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067) | Protein fusion: mid-exon (EML4-RET)	NA
EVT-P-0014808-T03-IM6-166	P-0014808-T03-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6 inv | Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-0014808-T02-IM6-167	P-0014808-T02-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6 inv | Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-0014808-T01-IM6-168	P-0014808-T01-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	12	713	True	lost		Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6 inv | Protein fusion: out of frame (RET-ERCC6)	NA
EVT-P-0045809-T02-IM6-169	P-0045809-T02-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622269	19	1063	True	retained	Protein kinase domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622269) | Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-0045809-T06-IM6-170	P-0045809-T06-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622323	19	1063	True	retained	Protein kinase domain			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622323) | Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-0025215-T01-IM6-171	P-0025215-T01-IM6		RET-GEMIN5	GEMIN5	in-frame	three_prime	43611067	11	712	True	retained	Protein kinase domain			GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067) | Protein Fusion: in frame {GEMIN5:RET}	NA
EVT-P-0027417-T01-IM6-173	P-0027417-T01-IM6		RET-KCNMA1	KCNMA1	out-of-frame	three_prime	43608158	9	550	True	retained	Protein kinase domain			KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627:KCNMA1_c.1649-143:RET | Protein Fusion: out of frame {KCNMA1:RET}	NA
EVT-P-0043704-T01-IM6-174	P-0043704-T01-IM6		RET-KIF5B	KIF5B	in-frame	five_prime	32306364	1	1	True	lost		Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv | Protein Fusion: in frame {RET:KIF5B}	NA

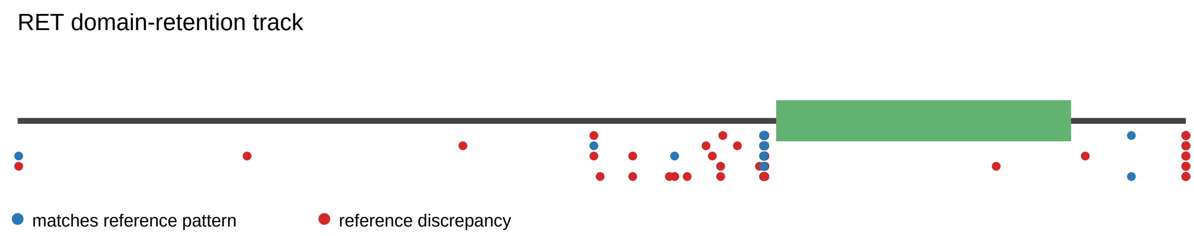
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006226-T02-IM5-178	P-0006226-T02-IM5		RET-NCOA4	NCOA4	in-frame	three_prime	43611893	12	713	True	retained	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0001151-T01-IM3-179	P-0001151-T01-IM3		RET-NCOA4	NCOA4	in-frame	three_prime	43611645	12	713	True	retained	Protein kinase domain			RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0009592-T01-IM5-180	P-0009592-T01-IM5		RET-NCOA4	NCOA4	unknown	three_prime	43611143	12	713	True	retained	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c.2136+621:RETdup | Protein fusion: mid-exon (NCOA4-RET)	NA
EVT-P-0001155-T01-IM3-181	P-0001155-T01-IM3		RET-NCOA4	NCOA4	in-frame	three_prime	43610941	11	712	True	retained	Protein kinase domain			RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0022818-T01-IM6-182	P-0022818-T01-IM6		RET-NCOA4	NCOA4	unknown	three_prime	43611746	12	713	True	retained	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0002145-T01-IM3-183	P-0002145-T01-IM3		RET-PDCD10	PDCD10	out-of-frame	five_prime	43609394	10	627	True	lost		Protein kinase domain		RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10,3)(q11.21;q26.1)(chr10:g.43609394::chr3:g.167419184) | Protein fusion: out of frame (RET-PDCD10)	NA
EVT-P-0001898-T01-IM3-184	P-0001898-T01-IM3		RET-RASSF4	RASSF4	in-frame	five_prime	43611749	12	713	True	lost		Protein kinase domain		RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del | Protein fusion: in frame (RET-RASSF4)	NA
EVT-P-0030396-T02-IM6-185	P-0030396-T02-IM6		RET-RBPMS	RBPMS	unknown	three_prime	43610018	11	657	False	retained	Protein kinase domain			RBPMS (NM_001008712) - RET (NM_020975) Rearrangement : t(8,10)(p21.1,q11.21)(chr8:g.30412656::chr10:g.43610018) | Protein Fusion: mid-exon {RBPMS:RET}	NA
EVT-P-0029222-T02-IM6-212	P-0029222-T02-IM6		RET-RUFY1	RUFY1	unknown	three_prime	43610037	11	663	False	retained	Protein kinase domain			RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5,10)(q35.3,q11.21)(chr5:g.179025890::chr10:g.43610037) | Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-0015310-T02-IM6-213	P-0015310-T02-IM6		RET-RUFY3	RUFY3	out-of-frame	three_prime	43610427	11	712	True	retained	Protein kinase domain			RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655316::chr10:g.43610427) | Protein fusion: out of frame (RUFY3-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015310-T04-IM6-214	P-0015310-T04-IM6		RET-RUFY3	RUFY3	unknown	three_prime	43610427	11	712	True	retained	Protein kinase domain			RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655251::chr10:g.43610427) | Protein Fusion: mid-exon {RUFY3:RET}	NA
EVT-P-0025609-T01-IM6-215	P-0025609-T01-IM6		RET-SHROOM3	SHROOM3	in-frame	three_prime	43611545	12	713	True	retained	Protein kinase domain			SHROOM3 (NM_020859) - RET (NM_020975) rearrangement: t(4;10)(q21.1;q11.21)(chr4:g.77664882::chr10:g.43611545) | Protein Fusion: in frame {SHROOM3:RET}	NA
EVT-P-0053530-T01-IM6-216	P-0053530-T01-IM6		RET-SLC4A4	SLC4A4	out-of-frame	five_prime	43610439	11	712	True	lost		Protein kinase domain		RET (NM_020975) - SLC4A4 (NM_001134742) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.72412310::chr10:g.43610439) | Protein Fusion: out of frame {RET:SLC4A4}	NA
EVT-P-0002145-T01-IM3-217	P-0002145-T01-IM3		RET-TFG	TFG	in-frame	three_prime	43609437	10	627	True	retained	Protein kinase domain			RET (NM_020975) - TFG (NM_001195478) translocation: t(10,3)(q11.21;q12.2)(chr10:g.43609437::chr3:g.100450751) | Protein fusion: in frame (TFG-RET)	NA
EVT-P-0021980-T01-IM6-218	P-0021980-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43610550	11	712	True	retained	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12-20): c.763-591:NCOA4_c.2136+366:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0017716-T01-IM6-219	P-0017716-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611172	12	713	True	retained	Protein kinase domain			NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0025531-T01-IM6-220	P-0025531-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611375	12	713	True	retained	Protein kinase domain			NCOA4 (NM_001145260) RET (NM_020975) fusion: c.619-108NCOA4_c.2137-657RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0034704-T01-IM6-222	P-0034704-T01-IM6		RET-TRIM24	TRIM24	in-frame	three_prime	43611827	12	713	True	retained	Protein kinase domain			TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::chr10:g.43611827) | Protein Fusion: in frame {TRIM24:RET}	NA
EVT-P-0046790-T01-IM6-223	P-0046790-T01-IM6		RET-UXS1	UXS1	unknown	five_prime	43600430	4	219	False	lost		Protein kinase domain		RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430) | Protein Fusion: mid-exon {RET:UXS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029222-T03-IM6-224	P-0029222-T03-IM6		RUFY1-RET	RUFY1	unknown	three_prime	179025890	20	1115	True	lost		Protein kinase domain		RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037) | Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-0031340-T02-IM6-225	P-0031340-T02-IM6		RUFY2-RET	RUFY2	in-frame	three_prime	70141796	20	1115	True	lost		Protein kinase domain		RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv | Protein Fusion: in frame {RUFY2:RET}	NA
EVT-P-0026614-T01-IM6-227	P-0026614-T01-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43611575	12	713	True	retained	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T06-IM6-228	P-0036400-T06-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T04-IM6-229	P-0036400-T04-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T03-IM6-230	P-0036400-T03-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	11	712	True	retained	Protein kinase domain			SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA

Figures

domain retention outliers



reference comparison